Resolution Geometry Closes Itself Where It Can

What the Rooting Theorem (Čencov + Hessian) Solves Inside RG: Chaos, Forgetting, Walls, Tame Rates, and the Exact Boundary of Self-Closure

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October 31, 2026 (print edition)

Abstract

Before anyone searches for exotic closures of RG's open problems, we determine how much RG's own rooting theorem already closes. The rooting theorem is Čencov's theorem derived from RG descent, together with the Hessian theorem. Every item is typed in one of four ways:

- CLOSED: proved here;
- CONDITIONAL: closed under one named, checkable hypothesis;
- REDUCED: equivalent to a named external problem;
- NOT REACHED: the machinery provably says nothing.

New structural results.

(i) *Local comparability.* Every smooth admissible divergence equals $\frac{f''(1)}{2}\chi^2$ to first relative order. Hence observational fibres, vanishing orders, Łojasiewicz exponents and wall rates (RLCTs) are invariants of the experiment, not of the chosen divergence.

(ii) *Forgetting.* The local forgetting rate of every admissible divergence is the same number, η_{χ^2}, the squared maximal correlation. Finite-scale rates form the band $\eta_{\chi^2} \le \eta_f \le \eta_{\mathrm{TV}}$.

(iii) *Lossless dynamics never forgets.* Invertible dynamics at full resolution is a lossless re-description, so chaos alone forgets nothing; forgetting is always a property of the protocol.

(iv) *Coarse-grained chaos is exact on Markov partitions.* At dyadic resolution k, the doubling map loses exactly one bit per step (Lyapunov $= \ln 2$) and forgets completely after k steps, while a rotation never forgets.

(v) *Chaos amplifies parameters.* The Fisher information about a dynamical parameter grows at most like $e^{2\lambda n}$, and the logistic map at $r = 4$ attains rate 1.37 against $2\ln 2 = 1.386$.

(vi) *Wall rates are rational and computable for tame models.* They are RLCTs, divergence-invariant: we find $\frac{1}{4}$ for θ^4, $\frac{1}{2}$ with multiplicity 2 for $\theta_1^2\theta_2^2$, and $\frac{1}{4}$ for both KL and χ^2 in a singular Bernoulli model.

Ledger.

- *L133* (forced turbulence) becomes CLOSED for stochastic forcing (import) and REDUCED to uniform mixing on declared observables for deterministic forcing.
- *L132* (tame types) becomes CLOSED for local rate types and stays NOT REACHED for the global enumeration.

- *Čencov's continuity question* is CLOSED for every metric arising as a Hessian.
- *Not reached:* nonlinear field equations, the Hamiltonian, the universal quantum recovery bound, quantum wall exponents (which are task-typed), and finite-panel certification.

Evidence. The battery is deterministic and exact where possible, with no random sampling.

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1 The rule: close from the root, or say exactly why not

RG's root is *descent*: an admissible quantity must be invariant under lossless re-descriptions. By the Čencov print, lossless re-descriptions are exactly congruent splittings. Descent therefore forces Fisher as the metric (Čencov print, Theorem 3.4) and as the Hessian of every admissible divergence (Hessian print, Theorem 3.1). This paper turns that root against RG's open items. It may add a hypothesis only if the hypothesis is named, checkable and not circular, i.e. not equivalent to the conclusion.

2 A new tool: local comparability

Lemma 2.1 (Local comparability)**.** *Let $f(1) = 0$, with f of class C^3 on $[1-\delta, 1+\delta]$, $f''(1) > 0$, and $M_3 = \sup |f'''|$ there. If $\max_i |p_i/q_i - 1| \le \delta$, then*

$$\left| D_f(p\|q) - \tfrac{f''(1)}{2}\chi^2(p\|q) \right| \;\le\; \tfrac{M_3\delta}{6}\,\chi^2(p\|q).$$

Proof. Put $r_i = p_i/q_i$. Taylor's formula gives

$$q_i f(r_i) = q_i\Big[f'(1)(r_i-1) + \tfrac{f''(1)}{2}(r_i-1)^2 + R_i\Big], \qquad |R_i| \le \tfrac{M_3}{6}|r_i-1|^3 \le \tfrac{M_3\delta}{6}(r_i-1)^2.$$

Sum over i. The linear term vanishes because $\sum_i q_i(r_i-1) = 0$, and $\sum_i q_i(r_i-1)^2 = \chi^2(p\|q)$. Symbolic check (X1): KL, squared Hellinger, χ^2 and Jensen–Shannon all have leading term $\frac{f''(1)}{2}\varepsilon^2$. □

Theorem 2.2 (Wall invariants belong to the experiment)**.** *Let $\theta \mapsto p_\theta$ be continuous into the positive laws near θ_0, and let D, D' be two smooth admissible f-divergences. Then:*

(a) the zero sets $\{\theta : D(p_\theta \| p_{\theta_0}) = 0\}$ *coincide, and equal the observational fibre;*

(b) near θ_0, $D \asymp D'$ *with ratio tending to* $f''(1)/\tilde{f}''(1)$, *where* $\tilde{f}$ *generates* D'. *Hence they share every Łojasiewicz exponent, and the vanishing order along every curve;*

(c) if the model is analytic, the real log canonical thresholds λ *and multiplicities* m *of* $K = D(p_\theta \| p_{\theta_0})$ *and* $K' = D'(p_\theta \| p_{\theta_0})$ *coincide.*

Proof. (a) Each D_f with f strictly convex vanishes iff $p = q$. (b) Apply Lemma 2.1 to both divergences, using continuity of $\theta \mapsto p_\theta / p_{\theta_0}$ to make $\delta \to 0$. (c) Write $Z_n(K) = \int e^{-nK} \varphi \, d\theta \sim C n^{-\lambda} (\log n)^{m-1}$. If $aK \le K' \le bK$ near θ_0, then $Z_{bn}(K) \le Z_n(K') \le Z_{an}(K)$. A constant rescaling of n changes neither λ nor m. □

Consequence. RG's walls (L129–L131) carry rate data, such as λ and the Łojasiewicz exponent, that no admissible choice of divergence can alter. These are genuine invariants of the protocol.

3 Forgetting and chaos

Theorem 3.1 (One local forgetting rate)**.** *For a channel* K *and an interior reference* p, *let*

$$\eta_D^{\mathrm{loc}}(K, p) = \sup_{u \neq 0} \frac{g^D_{Kp}(Ku)}{g^D_p(u)}.$$

Every admissible divergence (Hessian print, Theorem 3.1) gives the same value,

$$\eta^{\mathrm{loc}}(K, p) = \sigma_2(B)^2, \qquad B = \mathrm{diag}(Kp)^{-1/2} K \, \mathrm{diag}(p)^{1/2},$$

the squared second singular value, i.e. the squared maximal correlation.

Proof. By the Hessian theorem $g^D = cF$ on both sides, so c cancels. Substitute $u = \mathrm{diag}(p)^{1/2} w$ with $w \perp \sqrt{p}$. Then $F_p(u) = \|w\|^2$ and $F_{Kp}(Ku) = \|Bw\|^2$. The top singular pair of B is $(\sqrt{p}, \sqrt{Kp})$ with singular value 1, so the supremum over $w \perp \sqrt{p}$ is σ_2^2. □

Imported Theorem 3.2 (The finite-scale band [1, 2])**.** *For finite-scale contraction coefficients,* $\eta_{\chi^2} \le \eta_f \le \eta_{\mathrm{TV}}$ *for strictly convex smooth* f.

Witnesses (X2), at the uniform reference.

- Binary symmetric channel with $\varepsilon = 0.1$: $\eta_{\chi^2} = \eta_{\mathrm{KL}} = 0.64$ and $\eta_{\mathrm{TV}} = 0.8$.
- Z-channel: $0.538 < 0.621 < 0.700$.

The local rate is one number. The finite rate is a band, and its member is task-declared: the same pattern as the quantum metric band.

Theorem 3.3 (Lossless dynamics never forgets)**.** *Let* Φ *be a bijection of a finite resolution space, or more generally a bimeasurable bijection preserving null sets. Then* $D(\Phi_* p \| \Phi_* q) = D(p \| q)$ *for every admissible divergence. Invertible dynamics at full resolution forgets nothing, however chaotic.*

Proof. Φ and Φ^{-1} are both channels, so monotonicity in both directions gives equality. In the finite case Φ is a permutation, which is a lossless re-description (Čencov print, Lemma 3.2). Witness (X6): the cat map on a 61×61 discrete torus is a permutation, and KL between pushed laws is unchanged. □

Consequence for RG. Forgetting (L53–L55) is never a property of an invertible law alone. It enters only through the *protocol*: coarse-graining, noise, or a non-injective map.

Theorem 3.4 (Exact coarse-grained chaos on Markov partitions)**.** *Let Φ be a full-branch expanding map with b branches, each of uniform slope b, so that the level-k partition into $N = b^k$ cells is Markov. Each cell maps onto b cells with weight $1/b$ each. Under a uniform prior on the initial cell, the mutual information between the initial cell and the cell at time n is*

$$I_n = (k-n)\log b \quad (n \le k), \qquad I_n = 0 \quad (n \ge k).$$

A rotation, i.e. a permutation of cells, keeps $I_n = k \log b$ forever.

Proof. After $n \le k$ steps an initial cell is spread uniformly over b^n cells. These sets partition the space b^n-to-one across initial cells. So $H(X_0 \mid X_n) = n \log b$, and $I_n = H(X_0) - H(X_0 \mid X_n) = (k-n)\log b$. A permutation is lossless (Theorem 3.3). Witness (X3), doubling map with $k = 6$: $I_n = 6, 5, 4, 3, 2, 1, 0, 0$ bits exactly, and the rotation gives $6, 6, 6, \ldots$ □

The loss rate $\log b$ is the Lyapunov exponent and the Kolmogorov–Sinai entropy (Pesin, imported). Chaos is *how fast a declared resolution forgets*, not a loss inside the law.

Theorem 3.5 (Chaos amplifies parameter information)**.** *Let $x_{n+1} = f_r(x_n)$ with $\partial_r f$ bounded, and observe x_n with additive noise σ. The Fisher information about r is s_n^2/σ^2, where $s_{n+1} = \partial_r f(x_n) + f_r'(x_n) s_n$. Hence*

$$\limsup_{n\to\infty} \tfrac{1}{2n} \log s_n^2 \;\le\; \lambda,$$

the Lyapunov exponent along the orbit.

Proof. Variation of constants gives

$$s_n = \sum_{j<n} \Big(\prod_{j<i<n} f'(x_i) \Big)\, \partial_r f(x_j).$$

Each product is $e^{(n-j)(\lambda + o(1))}$, and the sum of a geometric series is dominated by its largest term. □

Witness (X4). For the logistic map at $r = 4$, in 300-digit arithmetic from a fixed initial point, the fitted rate of $\log s_n^2$ is 1.370, against $2\lambda = 2\ln 2 = 1.386$. The two faces of chaos are one Fisher calculus: information about the *state* at a fixed resolution contracts (Theorem 3.4), while information about *parameters* is amplified (Theorem 3.5).

Theorem 3.6 (L133 closed for stochastic forcing, reduced for deterministic forcing)**.** *Declare a resolution protocol, i.e. a class $\mathcal{O}$ of observables read from the state. Uniform forgetting of the initial law, as seen through $\mathcal{O}$, holds uniformly in a forcing parameter iff the dynamics is uniformly mixing on $\mathcal{O}$. By Theorem 3.1 the local rate is the same for all admissible divergences.*

- Stochastic forcing. *For the 2D Navier–Stokes equations with degenerate stochastic forcing, unique ergodicity and a spectral gap in a Wasserstein distance hold [3, 4], imported. Uniform forgetting on Lipschitz observables follows:* CLOSED.
- Deterministic forcing. *The condition is uniform mixing of the deterministic attractor on $\mathcal{O}$, an open problem of turbulence theory:* REDUCED.

Proof. Forgetting through $\mathcal{O}$ is decay of $|\mathbb{E}_p[o(X_n)] - \mathbb{E}_q[o(X_n)]|$ uniformly over o in the unit ball of $\mathcal{O}$. By duality this is the contraction of the law difference in the dual norm. By Theorem 3.3 it cannot come from the invertible part of the dynamics, so it is exactly mixing on $\mathcal{O}$. The imports supply the stochastic case. □

4 Walls and tame rates

Imported Theorem 4.1 (Singular learning [5]). *For an analytic model with the true law inside it, the Bayes free energy satisfies* $F_n = nS_n + \lambda \log n - (m-1)\log\log n + O_p(1)$. *Here* λ *is the RLCT of* $K(\theta) = \mathrm{KL}(p_{\theta_0} \| p_\theta)$, *it is rational, and it is computable by resolution of singularities. Regular points have* $\lambda = d/2$.

Theorem 4.2 (Wall rates are tame invariants). *For analytic (tame) models, the rate at which a wall can be resolved by data is the pair* (λ, m). *By Theorem 2.2(c) this pair is the same for every admissible smooth divergence. In particular a wall's rate type is an RG invariant computed from the experiment.*

Witnesses (X5), by deterministic quadrature over $n = 10^3, \ldots, 10^7$.

Germ	fitted λ	expected
θ^4	0.2500	1/4
$\theta_1^2 + \theta_2^2$ (regular)	1.0000	1
$\theta_1^2\theta_2^2$	0.522	1/2, with $m = 2$
singular Bernoulli, KL	0.2500	1/4
singular Bernoulli, χ^2	0.2500	1/4

Meaning. A wall where Fisher degenerates is not merely "unidentifiable". Its data-resolution rate $n^{-\lambda}$ is fixed, rational and divergence-invariant. This is the first quantitative typing of RG walls by the rooting theorem.

No-Go 4.3 (What the rate type does not give). The right-equivalence type of the divergence germ, for example an ADE class [6], is *not* divergence-invariant. Lemma 2.1 controls only the order and the zero set, not higher jets. A finer typing is therefore task-declared. The *global enumeration* of tame characteristic types (L132), a counting problem of Hilbert-16 type, is not reached by any local theorem.

5 Čencov's continuity question, closed for Hessians

Proposition 5.1 (Continuity is automatic for Hessian metrics). *If an RG metric is the Hessian of an admissible divergence that is jointly* C^2 *near the diagonal, then the continuity hypothesis (R2) of the Čencov print holds automatically, and the metric is* c*·Fisher. Every* f*-divergence with* $f \in C^2$ *is of this kind. The open problem of the Čencov print therefore survives only for metrics that arise from no smooth divergence.*

Proof. The Hessian $p \mapsto \partial_t^2 D(p + tu, p)|_0$ of a jointly C^2 function is continuous. Then apply the Hessian print, Theorem 3.1. □

6 Where the root provably stops

No-Go 6.1 (Not reached).

- *Nonlinear field equations.* The root fixes information geometry, not dynamical laws. The cubic gravity vertex needed gauge consistency, not descent (GR print).
- *The Hamiltonian.* Every Hermitian H passes every gate (QM print).
- *A universal quantum recovery bound for the plain Petz map.* Descent gives a band, not a recovery map.
- *Quantum wall exponents.* Different Petz metrics have *different* wall exponents (BKM grows logarithmically, RLD as $1/\varepsilon$). Quantum walls are therefore task-typed, unlike classical walls (Theorem 2.2).

- *Finite-panel certification.* No finite panel certifies descent without a declared modulus (Čencov print, No-Go 4.9).
- *Global tame enumeration* (L132) and *deterministic mixing* (L133, deterministic forcing).

7 Closure ledger

Item	Status	Added hypothesis	Evidence
Divergence-independence of walls (fibres, orders, λ)	CLOSED	smooth f, $f''(1) > 0$	Lemma 2.1, Theorem 2.2, X1, X5
Local forgetting rate (L53–L55)	CLOSED	—	Theorem 3.1, X2
Finite forgetting rate	band, task-declared	choice of f	imported band, X2
Chaos versus forgetting	CLOSED	—	Theorems 3.3, 3.4, X3, X6
Chaos amplifies parameters	CLOSED (upper bound)	bounded $\partial_r f$	Theorem 3.5, X4
L133, stochastic forcing	CLOSED	noise nondegeneracy (imported)	Theorem 3.6
L133, deterministic forcing	REDUCED	uniform mixing on $\mathcal{O}$	Theorem 3.6
Wall rate type (tame)	CLOSED	analyticity	Theorem 4.2, X5
L132 global enumeration	NOT REACHED	—	No-Go 4.3
Čencov continuity	CLOSED for Hessians	joint C^2	Proposition 5.1
Nonlinear GR, Hamiltonian, Petz bound, quantum walls, finite panels	NOT REACHED	—	No-Go 6.1

Answer to the question "how much can the root close?"

- Everything that is a *question about information geometry* closes: walls, forgetting, the chaos/protocol split, rates and continuity. Each item is proved here or reduced to one named external hypothesis.
- Everything that is a *question about which dynamical law nature uses* is provably outside the root: field equations, Hamiltonians and deterministic mixing.

Searching for an exotic closure of the first kind is unnecessary. For the second kind it is necessary, and it must come from outside RG.

8 Battery and limits

- `rg_rooting_closure.py` runs X1–X6. It is deterministic, with no random sampling, and reruns are byte-identical. It uses symbolic series (X1), exact Markov partitions (X3), 300-digit arithmetic (X4), deterministic quadrature (X5) and exact integer permutations (X6).
- *Limits.*

– The contraction coefficients in X2 are evaluated at the uniform reference; the KL value uses a 4001-point grid.
– X4 uses one initial point; the theorem gives the upper bound, and equality is generic.
– The λ fit for the product germ carries a multiplicity correction, so 0.522 approximates $\frac{1}{2}$.
– The imported results are Pesin, Watanabe, Hairer–Mattingly and the contraction band.

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